

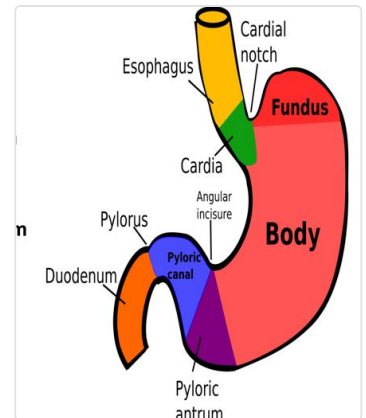
**BEFORE YOU READ – 3 BLIND ID CHALLENGES**

1. Name the four gastric regions — which two are histologically identical?
2. Which gland cells sit in the neck and which in the base — and what does each secrete?
3. How do pits and glands differ in the cardiac, fundic and pyloric zones?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 · Regions and wall — one-line ID rules

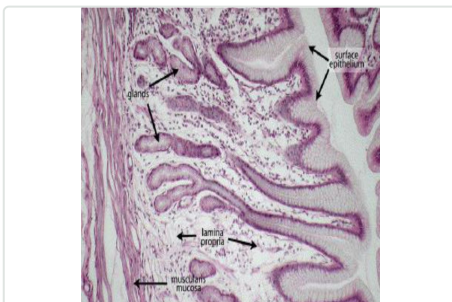
- **Four regions:** cardia (from the squamocolumnar junction), fundus (dome under the diaphragm), body, pylorus (continuous with the duodenum through the pyloric sphincter). **Fundus and body are histologically identical.**
- **Rugae:** longitudinal folds of mucosa + submucosa in the undistended stomach; they flatten when it fills.
- Same four tunics as the rest of the tract. **Epithelium:** simple columnar, dipping into the lamina propria as **gastric pits**; **every** surface and pit cell secretes **mucus**, forming a thick layer that protects against acid, with tight junctions as part of the barrier.



Cardia, fundus, body, pylorus.

DISCRIMINATOR GRID – THE THREE GLAND ZONES (THE ONE FINDING: PIT DEPTH VS GLAND LENGTH)

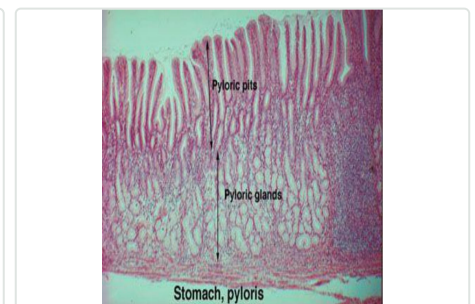
	CARDIAC	FUNDUS / BODY	PYLORIC
Where	small area near the esophageal opening	largest region, cardia to pylorus	distal 4-5 cm , leading to the small intestine
Glands	branched tubular cardiac glands	branched tubular gastric (fundic) glands ; each pit receives several	pyloric glands
Pits : glands	shallow pits , glands of equal length (slide wording)	shallow pits, long glands	long pits, short glands
Cells	mostly mucus -secreting; few parietal	all the gland cell types	mucus cells predominate; chief cells scarce; G cells (gastrin), D cells (somatostatin)



Cardiac: surface epithelium, glands, lamina propria, muscularis mucosa.



Fundus / body: gastric pits, long gastric glands, muscularis mucosae.



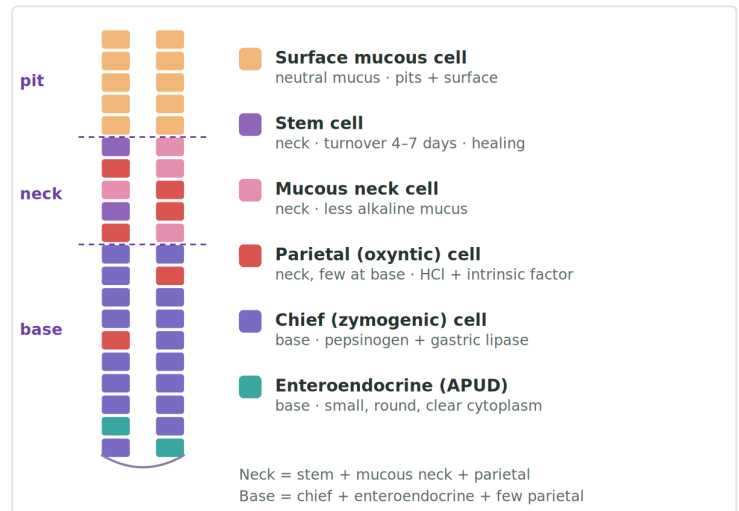
Pylorus: long pyloric pits, short pyloric glands.

- **Lamina propria:** loose CT with smooth muscle, lymphoid cells and glands. **Muscularis mucosae:** smooth muscle producing local movements of the mucosa.
- **Muscularis externa:** smooth muscle in **three directions** — **outer longitudinal, middle circular, inner oblique**; at the pylorus the **middle layer thickens** into the **pyloric sphincter**. **Serosa:** thin, covered by mesothelium.

2 · Gland cells — anchor diagram and grid

- Gland zones run **pit** → **neck** → **base**, and the cell mix is **not uniform**.
- Neck:** stem cells, mucous neck cells, parietal (oxyntic) cells.
- Base:** a few parietal cells, chief (zymogenic) cells, enteroendocrine cells.

Look-alike pair: parietal vs chief. **Parietal** = **pale, round, acidophilic, central nuclei, mitochondria (HCl + intrinsic factor); chief** = **basophilic, RER-rich, pepsinogen**. Parietal cells crowd the neck; chief cells dominate the base.

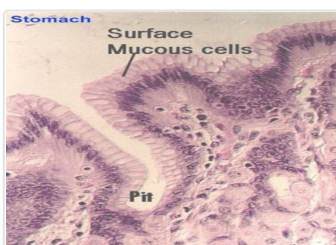


Anchor: one gastric gland, pit at the top.

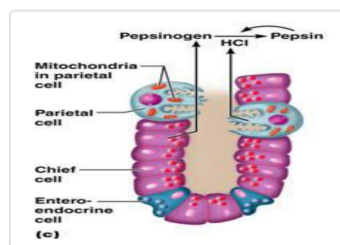
CELL	ID ON THE SLIDE	PRODUCT / JOB
Surface mucous	tall columnar, basal nuclei, clear cytoplasm with small mucin vacuoles , short microvilli	neutral mucus protecting the surface from acid
Stem	low columnar, oval nuclei near the base; high mitotic rate	turnover 4-7 days ; increase in number when the epithelium is damaged (e.g. gastric ulcer) → healing
Mucous neck	irregular, nucleus at the base, granules near the apex; clusters or single between parietal cells	less alkaline mucus than surface cells
Parietal (oxyntic)	large, pale, round-pyramidal ; 1-2 central nuclei; acidophilic cytoplasm, many mitochondria	HCl; intrinsic factor (glycoprotein needed for vitamin B12 absorption)
Chief (zymogenic)	smaller than parietal; basophilic (much RER, ribosomes); pepsinogen granules	pepsinogen → pepsin in stomach acid (starts protein hydrolysis); gastric lipase
Enteroendocrine (APUD)	small, round, clear cytoplasm, neurosecretory granules	gastrin, endorphins, serotonin (see note below)

MAGNIFICATION LADDER (SCAN-PATH ORGANISATION OF THE LECTURE'S FEATURES – THE LECTURE GIVES NO MAGNIFICATIONS)

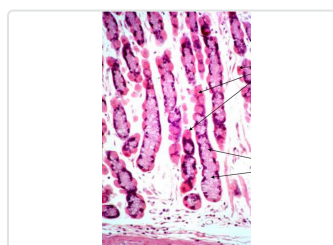
MAG	WHAT TO LOOK FOR
×4	Mucosa from surface to muscularis mucosae: pits on top, glands below — the pit : gland ratio names the region (cardiac / fundic / pyloric). Rugae fold the mucosa + submucosa.
×10	Gland architecture: neck (stem, mucous neck, parietal) vs base (chief, enteroendocrine, few parietal); several glands open into each pit.
×40	Cell identity: pale round parietal cells vs basophilic chief cells; surface mucous cells with clear mucin-filled cytoplasm.



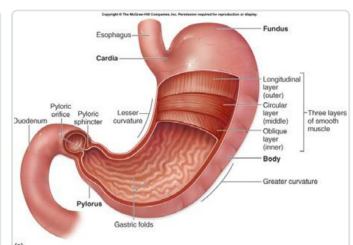
Surface mucous cells lining a pit.



Parietal, mucous neck and chief cells (H&E / PAS).



Gland: parietal and chief cells.



Three muscle layers; gastric folds; pyloric sphincter.

Slide inconsistencies: (1) the fundic enteroendocrine cell is listed as secreting gastrin, but the APUD table and the pylorus slide place gastrin's **G cells in the pylorus**; (2) diagrams describe surface / neck mucus as "alkaline" or "acidic" while the text says "neutral" and "less alkaline" — use the text wording.

3 · APUD (enteroendocrine) cells of the GIT — the lecture's table

CELL	SITE	PRODUCT	ACTION (as the lecture gives it)
D	pylorus, duodenum, pancreatic islets	somatostatin	—
EC	stomach, small + large intestine	serotonin, substance P	↑ gut motility
G	pylorus	gastrin	—
I	small intestine	cholecystokinin (CCK)	pancreatic enzyme secretion, gallbladder contraction
K	duodenum, jejunum	gastric inhibitory polypeptide (GIP)	—
L	ileum, colon	glucagon-like peptide; peptide YY	GLP → insulin secretion , sense of hunger; PYY → water + electrolyte absorption in the large intestine
Mo	small intestine	motilin	↑ gut motility
N	ileum	neurotensin	—
S	small intestine	secretin	pancreatic + biliary bicarbonate and water

The lecture also lists "gastric acid secretion" beside G, I, K, L, N and S cells without saying whether it rises or falls — do not answer the direction from this table.

Pearls — what must survive if everything else is forgotten

- **Four regions** (cardia, fundus, body, pylorus); **fundus = body** histologically; rugae flatten when the stomach fills.
- **Pits : glands** — cardiac: shallow pits; fundus / body: shallow pits + **long** glands; pylorus: **long** pits + **short** glands (mucus cells dominate; G and D cells).
- **Neck** = stem + mucous neck + parietal; **base** = chief + enteroendocrine + few parietal.
- **Parietal** = HCl + **intrinsic factor** (vitamin B12). **Chief** = **pepsinogen** (→ pepsin in acid) + gastric lipase. **Stem cells** = 4–7 day turnover, heal ulcers.
- **Muscularis**: outer longitudinal, middle circular, inner oblique; the middle layer thickens into the **pyloric sphincter**.

Test yourself — transfer-task MCQs

Unseen fields, not slide recall. Commit to an answer before you uncover the key.

- 1. A patient loses all parietal cells. Which vitamin absorption is impaired, and which product is missing?**
 - A. Vitamin B12; pepsinogen
 - B. Vitamin B12; intrinsic factor
 - C. Vitamin C; gastrin
 - D. Folate; gastric lipase
 - E. Vitamin K; mucus
- 2. A stomach biopsy shows long pits, short glands, predominantly mucus-secreting cells, scarce chief cells and scattered G cells. Which region?**
 - A. Cardia
 - B. Fundus
 - C. Body
 - D. Pylorus
 - E. Duodenum
- 3. Following a gastric ulcer, which cell population expands to heal the epithelium, and where is it found?**
 - A. Chief cells at the base of the glands
 - B. Stem cells in the neck region
 - C. Parietal cells in the pit
 - D. Enteroendocrine cells at the base
 - E. Surface mucous cells in the muscularis mucosae
- 4. A basophilic cell at the base of a fundic gland is packed with RER and membrane-limited granules of an inactive enzyme. What happens to the granule content in the stomach lumen?**
 - A. It is converted by acid into pepsin
 - B. It is converted by intrinsic factor into gastrin
 - C. It is absorbed unchanged
 - D. It becomes mucin
 - E. It is converted to HCl

5. A section through the pyloric region shows a thick ring of muscle. Which layer thickens to form the sphincter?

- A.** Outer longitudinal
- B.** Middle circular
- C.** Inner oblique
- D.** Muscularis mucosae
- E.** Serosa

ANSWERS – COVER UNTIL YOU'VE COMMITTED

- 1. B** — Parietal cells secrete HCl and intrinsic factor, a glycoprotein required for vitamin B12 absorption.
- 2. D** — Long pits, short glands, mucus cells predominant, scarce chief cells, G and D cells = pyloric region.
- 3. B** — Stem cells in the neck have a high mitotic rate (turnover 4-7 days) and increase when the epithelium is damaged.
- 4. A** — Chief cell pepsinogen is converted in the acid of the stomach into active pepsin, which starts protein hydrolysis.
- 5. B** — At the pylorus the middle circular layer of the muscularis thickens into the pyloric sphincter.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Histology GIT Block · MEDucation by Mattin Majid · Source: GIT Histology LG 2, Dr Hero K. Mustafa
Study alongside the lecture, not instead of it.